

Operating the n-adic Collatz Laboratory

— A Practical Guide for Simulating Numerical Universes —

This guide explains how to use the Python-based “**n_adic_lab**” to simulate, observe, and scan generalized Collatz trajectories.

1. Environment Setup

All you need is a Python environment (version 3.7 or higher). You can run the code by manually typing or pasting it into your Python interpreter or notebook.

2. Universe Configuration Steps

Step A: Defining the Base (n)

Upon launch, the system prompts you to enter a base value n .

- **Composite bases** (e.g., 4, 6, 12): Provide strong “gravitational pull” through multiple divisors
- **Prime bases** (e.g., 3, 5, 7): Create high-energy universes with fewer braking points

Step B: Setting Jump Constants (c_r)

For each remainder r that is coprime to n , you must define a jump constant c_r .

- **Golden Rule:** $r + c_r$ must be divisible by n (i.e., $r + c_r \equiv 0 \pmod{n}$)
- **Design Tip:** Setting $c_r = -r$ creates a fixed point at that residue

3. Choosing an Observation Mode

Mode 1: Single Orbit Investigation

Enter any positive integer x . The lab will trace its trajectory step by step.

- **Convergence:** The number reaches 1 (the “ground state”)
- **Cycle / Fixed Point:** The number enters a repeating loop
- **Divergence:** The number grows beyond 10^{18}

Note: Not all numbers necessarily converge to 1. Cycles and divergence are possible depending on the universe.

Mode 2: Universal Range Scan (1–100)

Automatically scans integers from 1 to 100 to map the “stability” of your universe.

- **Convergence Rate:** Percentage of numbers that reach 1
- **Attractor Discovery:** Detect hidden cycles or fixed points

4. Example Experiment: The “4-adic Trap”

To replicate the discovery of the 4-node loop involving 11:

1. Set base $n = 4$
2. Set $c_1 = -1$
3. Set $c_3 = 1$
4. Run Mode 2 (Scan)
5. Observation: About 30% of numbers fall into the loop $7 \rightarrow 9 \rightarrow 11 \rightarrow 14 \rightarrow 7$ instead of reaching 1

5. Troubleshooting

- **“Inconsistency” Error:** Your c_r does not satisfy $r + c_r \equiv 0 \pmod n$. Recalculate accordingly.
- **Infinite Loop:** If a number neither converges nor cycles within 5,000 steps, the system halts to protect your machine.

6. Citation & Contribution

If you discover a unique universe or an exceptional orbit, feel free to share your parameters (n and c_r values) and cite this work under the CC BY 4.0 License.

Next Steps

With the Overview (the “What”) and this Operation Guide (the “How”), your Collatz Universe Kit is complete.

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